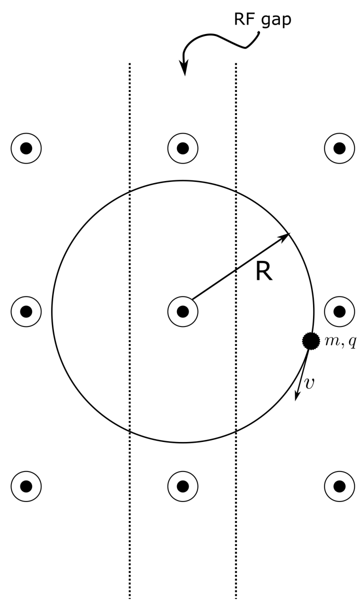
Instructions:

- Solutions to the problem sets are due by 9:00 a.m. the next morning.
- While points may differ between Problem Sets each Problem Set will carry equal weight for course grade.
- Please list your collaborators on your submission.

Problem 1. Cyclotron - Single Particle 20 pts.



a) 5 pts: Neglecting the change in orbit radius, R, over one transit, follow the steps in class to show that the non-relativistic period τ of the orbit is:

$$\tau = \frac{2\pi}{\omega_c}$$

where $\omega_c = \frac{qB}{m}$ is the cyclotron frequency and is constant.

Discuss the implications for a particle entering the oscillating RF field gap each transit. Will the frequency of the RF field need to change?

b) 5 pts: If a particle with mass, m, is traveling relativistically with approximately constant velocity, v, over one turn in a cyclotron, show that:

$$\frac{d\vec{p}}{dt} = -\frac{m\gamma v^2}{R}\hat{r}$$

where $\vec{p} = \gamma m\vec{v}$ and $\gamma = 1/\left(1 - \frac{\vec{v}^2}{c^2}\right)^{\frac{1}{2}} \approx \text{const.}$

c) 10 pts: Derive a relativistically correct formula for the period, τ, to characterize how it varies with particle kinetic energy, $\mathcal{E} = (\gamma - 1)\, mc^2$.

Discuss the implications of this result for the operation of a cyclotron as the particle becomes relativistic.

SOLUTION: Cyclotron - Single Particle 20 pts.

a) 5 pts: Balance the centrifugal and Lorentz forces then plug in equation for τ to remove R/v.

$$\tau = \frac{2\pi r}{v}$$
$$qvB = \frac{mv^2}{r} \implies \frac{r}{v} = \frac{m}{qB}$$
$$\tau = \frac{2\pi m}{qB} \equiv \frac{2\pi}{\omega_c} \implies \omega_c = \frac{qB}{m}$$

In the non-relativistic approximation, the cyclotron frequency is independent of the energy of the particle. The charge, mass, and static vertical B-field are all constants, so the cyclotron frequency is a constant as well. The RF frequency will not need to change, in this approximation.

b) 5 pts:

$$\mathbf{p} = \gamma m \mathbf{v} \qquad \gamma = \left(1 - \frac{\mathbf{v}^2}{c^2}\right)^{-1/2} \simeq \text{const.}$$
$$\frac{d\mathbf{p}}{dt} \approx \gamma m \frac{d\mathbf{v}}{dt} = \gamma m \frac{d\mathbf{v}}{d\phi}\frac{d\phi}{dt} = \gamma m \omega \frac{d\mathbf{v}}{d\phi}$$
$$\omega = \frac{v}{R} \qquad \frac{d\mathbf{v}}{d\phi} = -v\hat{r}$$
$$\frac{d\mathbf{p}}{dt} = -\gamma m \frac{v^2}{R}\hat{r}$$

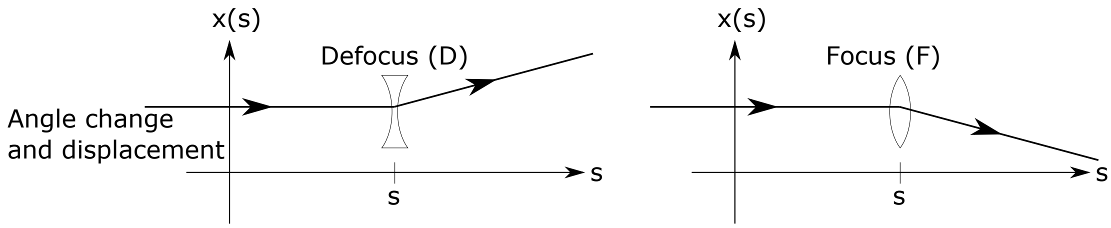
c) 10 pts: Follow the same steps as part (a), but use $\dot{\mathbf{p}}$ from part (b).

$$qvB = \frac{\gamma m v^2}{R} \implies \frac{v}{R} \equiv \omega_c = \frac{qB}{\gamma m} = \frac{qBc^2}{E_{tot}}$$
$$\tau = \frac{2\pi}{\omega_c} = \frac{2\pi E_{tot}}{qBc^2} = \frac{2\pi(\mathcal{E} + mc^2)}{qBc^2}$$
$$\tau = \frac{2\pi}{\omega_c}\left(1 + \frac{\mathcal{E}}{mc^2}\right)$$

The cyclotron period varies linearly with the kinetic energy of the particle ($\mathcal{E}$). Therefore, the frequency of the RF must change in order for the resonance condition to remain satisfied. Therefore constant frequency cyclotrons won't work for very relativistic beams or continuous bunch trains. Other techniques, such as varying the frequency must be used instead.

Problem 2. Thin Lens FODO Phase Advance and Stability 15 points

A thin lens changes the angle of a particle trajectory but not the coordinate:



This action can be specified by transfer matrices applied at $s = s_i$

$$\begin{bmatrix} x \\ x' \end{bmatrix} = \mathbf{M}(s|s_i) \begin{bmatrix} x_i \\ x_i' \end{bmatrix}$$

Focusing:

$$\mathbf{M}_F = \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix} \qquad f > 0$$

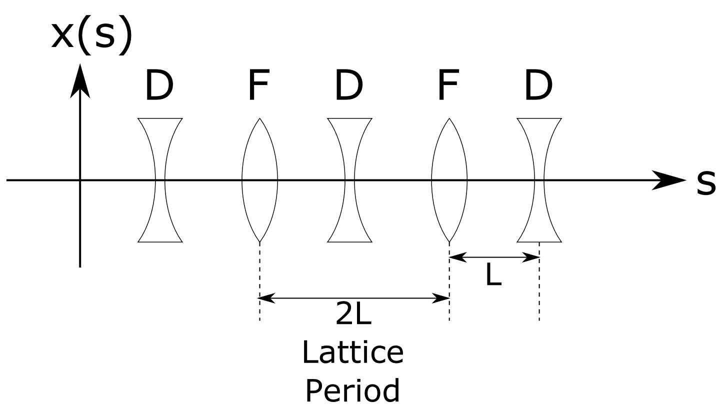
Defocusing:

$$\mathbf{M}_D = \begin{bmatrix} 1 & 0 \\ \frac{1}{f} & 1 \end{bmatrix} \qquad f > 0$$

A free space drift of length L has a transport matrix:

$$\mathbf{M}_0 = \begin{bmatrix} 1 & L \\ 0 & 1 \end{bmatrix}$$

Consider a lattice of period $2L$ made up of equally spaced F and D lenses with equal values of f.

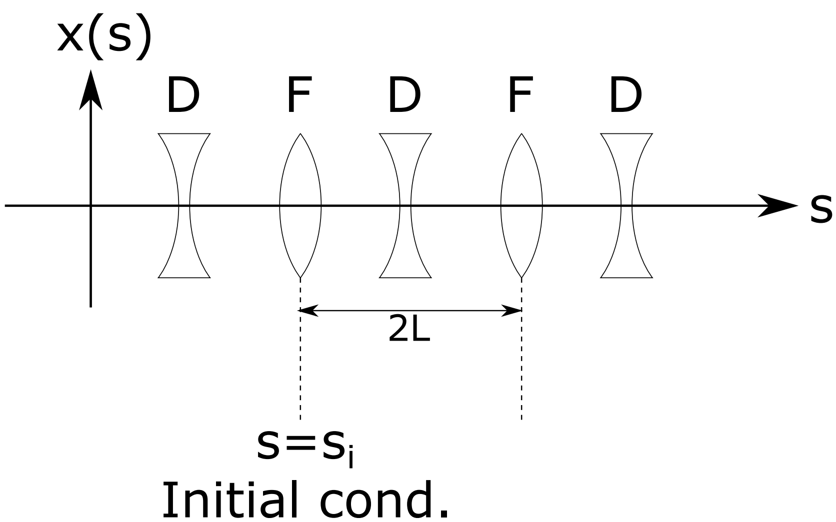


a) 10 pts: Calculate $\cos\Phi$ where Φ is the particle phase advance. Express the answer in terms of L/f. Also, determine the range of f for which the particle orbit is stable.

b) 5 pts: For the case of f chosen to correspond to the stability limit, sketch the motion of a particle with initial conditions:

$$\lim_{s \to s_i} x(s) = x_0$$
$$\lim_{s \to s_i} x'(s) = x_o/L$$

where $s = s_i$ is the axial location of a focusing thin lens kick, and $s \to s_i^-$ is just before the kick. Sketch the particle orbit for focusing strength slightly off from the stability limit into the unstable region. Superimpose the orbit sketch on a diagram of the lattice (see below):



a) 10 pts: $\mathbf{M} = \mathbf{M}_F \cdot \mathbf{M}_O \cdot \mathbf{M}_D \cdot \mathbf{M}_O$.

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & L \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ \frac{1}{f} & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & L \\ 0 & 1 \end{bmatrix}$$
$$\mathbf{M} = \begin{bmatrix} 1 + \frac{L}{f} & 2L + \frac{L^2}{f} \\ -\frac{L}{f^2} & 1 + \frac{L}{f} - \frac{L^2}{f^2} \end{bmatrix}$$

or equivalently, you can write:

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2f} & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & L \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ \frac{1}{f} & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & L \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ -\frac{1}{2f} & 1 \end{bmatrix}$$
$$\mathbf{M} = \begin{bmatrix} \frac{L+f}{f} & \frac{L(L+2f)}{f} \\ -\frac{L}{f^2} & -\frac{L^2}{f^2} - \frac{L}{f} + 1 \end{bmatrix}$$

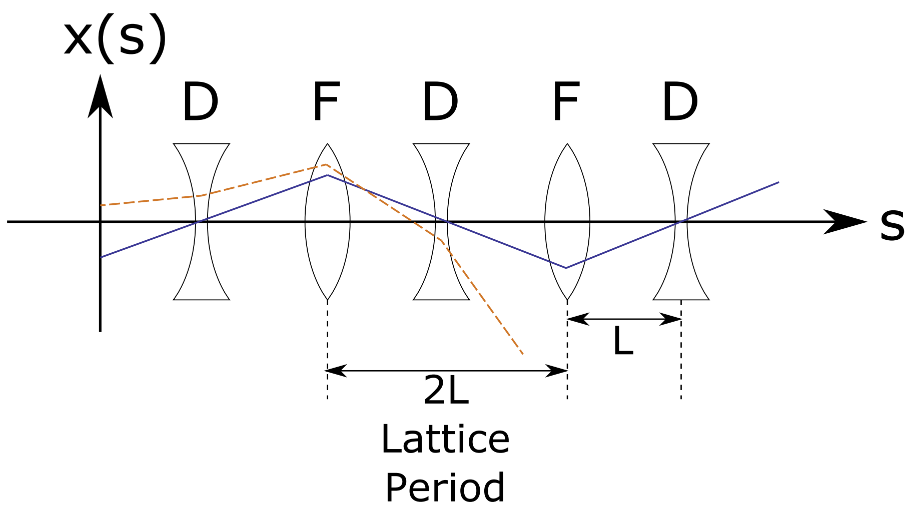
For both:

$$\implies \cos\Phi \equiv \frac{1}{2}Tr[\mathbf{M}] = 1 - \frac{L^2}{2f^2}$$

For stability:

$$\cos\Phi = \frac{1}{2}\mathrm{Tr}[\mathbf{M}] \leq 1$$
$$\left|1 - \frac{L^2}{2f^2}\right| \leq 1$$
$$\implies 0 \leq \frac{L}{f} \leq 2$$

b) 10 pts: The phase advance of the particle "oscillations" advances by π each lattice period (axial length $2L$). The orbit is a straight line between drifts. At the stability limit, the amplitude of the maximum orbit will not change. Beyond the stability limit the maximum orbit will increase



Solid line - at stability limit.

Dashed line - Beyond stability limit.

Note: At the stability limit, $x = 0$ in each D lens and x will alternate between $x = \pm x_0$ at each F lens. Beyond the stability limit, it is clear that the kicks will amplify the particle amplitude, and the particle trajectory increases in amplitude while traversing sucessive lenses.

Problem 3. Drift Twiss Parameters 20 pts.

Consider a long, field-free region adjacent to a collision point. The lattice is designed such that the beam size is minimized at the collision point, represented by $\beta = \beta^*$. Recall that the Twiss functions are propagated from s to s_0 using the transfer matrix:

$$\begin{bmatrix}\beta\\ \alpha\\ \gamma\end{bmatrix} = \begin{bmatrix} \mathrm{m}_{11}^2 & -2m_{11}m_{12} & m_{12}^2 \\ -m_{21}m_{11} & m_{11}m_{22}+m_{12}m_{21} & -m_{12}m_{22} \\ m_{21}^2 & -2m_{22}m_{21} & m_{22}^2 \end{bmatrix}\begin{bmatrix}\beta_0\\ \alpha_0\\ \gamma_0\end{bmatrix}$$

a) 5 pts: Use the transfer matrix propagation to show that the β-function evolves with s like

$$\beta(s) = \beta^* + \frac{(s - s_0)^2}{\beta^*}$$

b) 5 pts: How does $\alpha(s)$ evolve?

c) 5 pts: How does the phase $\psi(s)$ evolve?

d) 5 pts: What is the largest phase advance possible across a field-free region?

(a) For this problem, it is important to note that at $\beta^\star$, $\alpha = 0$ (the beam waist). Using equation the transfer matrix, we get

$$\beta = m_{11}^2\beta^\star + m_{12}^2\gamma$$

Since $\alpha = 0$, $\gamma = 1/\beta^\star$ and this simplifies to

$$\beta = m_{11}^2\beta^\star + \frac{m_{12}^2}{\beta^\star}$$

This is a drift case, so $m_{11} = 1$ and $m_{12} = \Delta s$. Thus

$$\boxed{\beta(s) = \beta^\star + \frac{(s - s_0)^2}{\beta^\star}}$$

(b) From the matrix, the equation for α is

$$\alpha = -m_{21}m_{11}\beta^\star - \frac{m_{12}m_{22}}{\beta^\star}$$

In this case $m_{21} = 0$, $m_{12} = \Delta s$ and $m_{11} = m_{22} = 1$ Thus

$$\boxed{\alpha(s) = -\frac{s - s_0}{\beta^\star}}$$

We see that α evolves linearly. We could also demonstrate that $\gamma(s)$ remains constant in a drift moving away from the local beam waist.

(c) We know

$$\frac{d\psi}{ds} = \frac{1}{\beta} \implies \phi(s) = \int \frac{ds}{\beta(s)}$$

$$\psi(s) = \int \frac{ds}{\beta^\star + \frac{(s-s_0)^2}{\beta^\star}}$$

Integrating and assuming $\psi(0) = 0$,

$$\boxed{\psi(s) = \arctan\left(\frac{s - s_0}{\beta^\star}\right)}$$

(d) Since

$$\arctan \to -\frac{\pi}{2} \text{ as } s \to -\infty$$

$$\arctan \to \frac{\pi}{2} \text{ as } s \to \infty$$

then the $\boxed{\text{maximum possible phase advance is } \pi}$ (180 degrees).